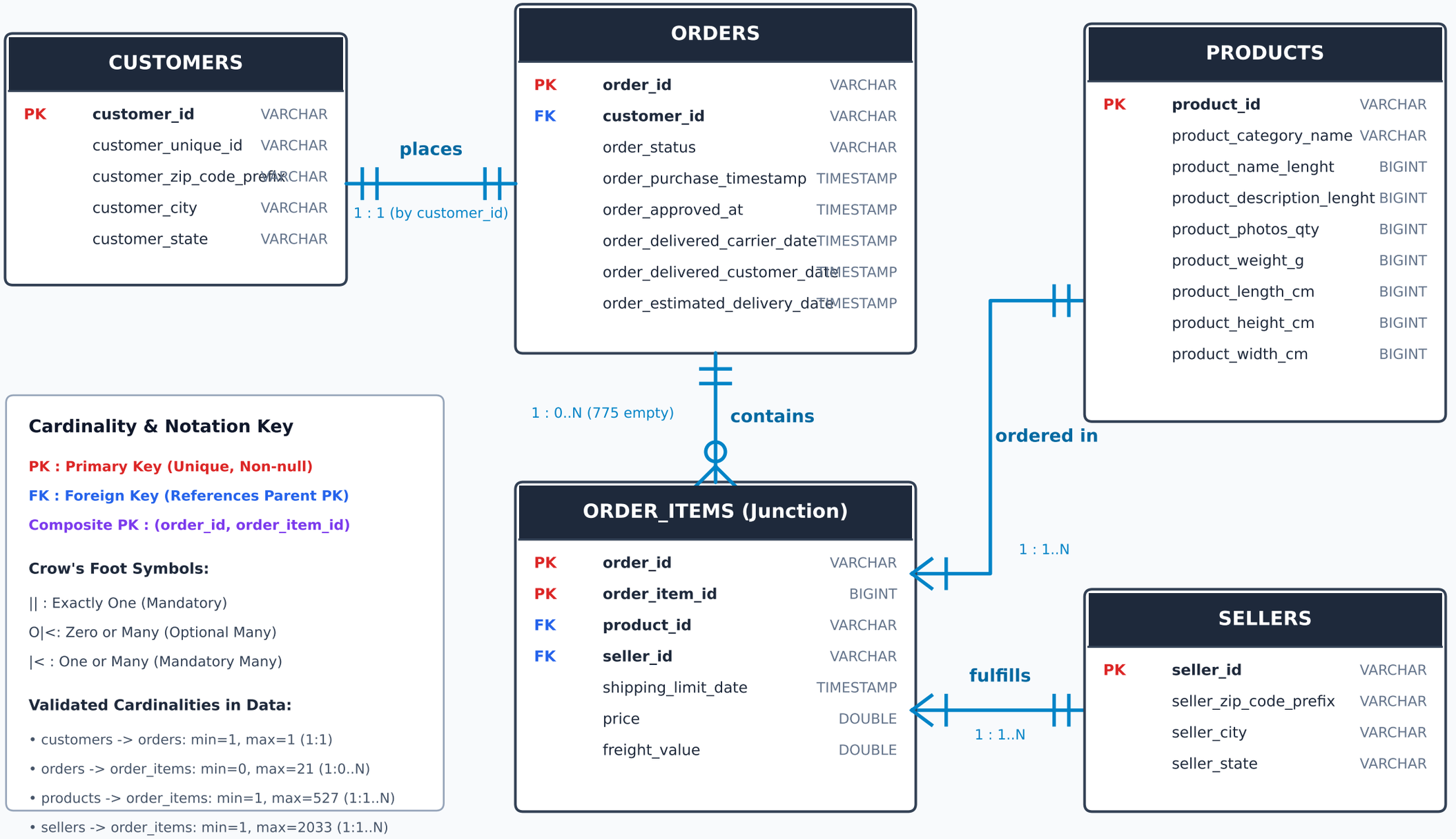
# Entity-Relationship Diagram (ERD) - Olist E-Commerce

Crow's Foot Notation | Primary Keys (PK) & Foreign Keys (FK) Validated via DuckDB




### Cardinality & Notation Key

**PK : Primary Key (Unique, Non-null)**

**FK : Foreign Key (References Parent PK)**

**Composite PK : (order_id, order_item_id)**

**Crow's Foot Symbols:**

|| : Exactly One (Mandatory)

O|<: Zero or Many (Optional Many)

|< : One or Many (Mandatory Many)

**Validated Cardinalities in Data:**

- customers -> orders: min=1, max=1 (1:1)
- orders -> order_items: min=0, max=21 (1:0..N)
- products -> order_items: min=1, max=527 (1:1..N)
- sellers -> order_items: min=1, max=2033 (1:1..N)